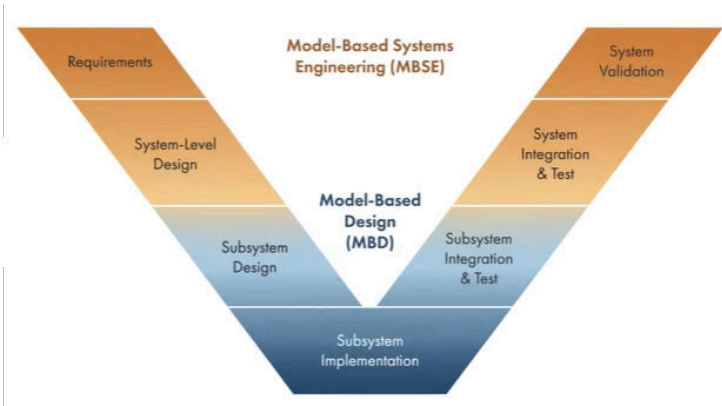


Simulink Fundamentals

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Simulink is the block-diagram environment for modeling and simulating dynamic systems. This guide uses the model workshopModel.slx that is part of this project.

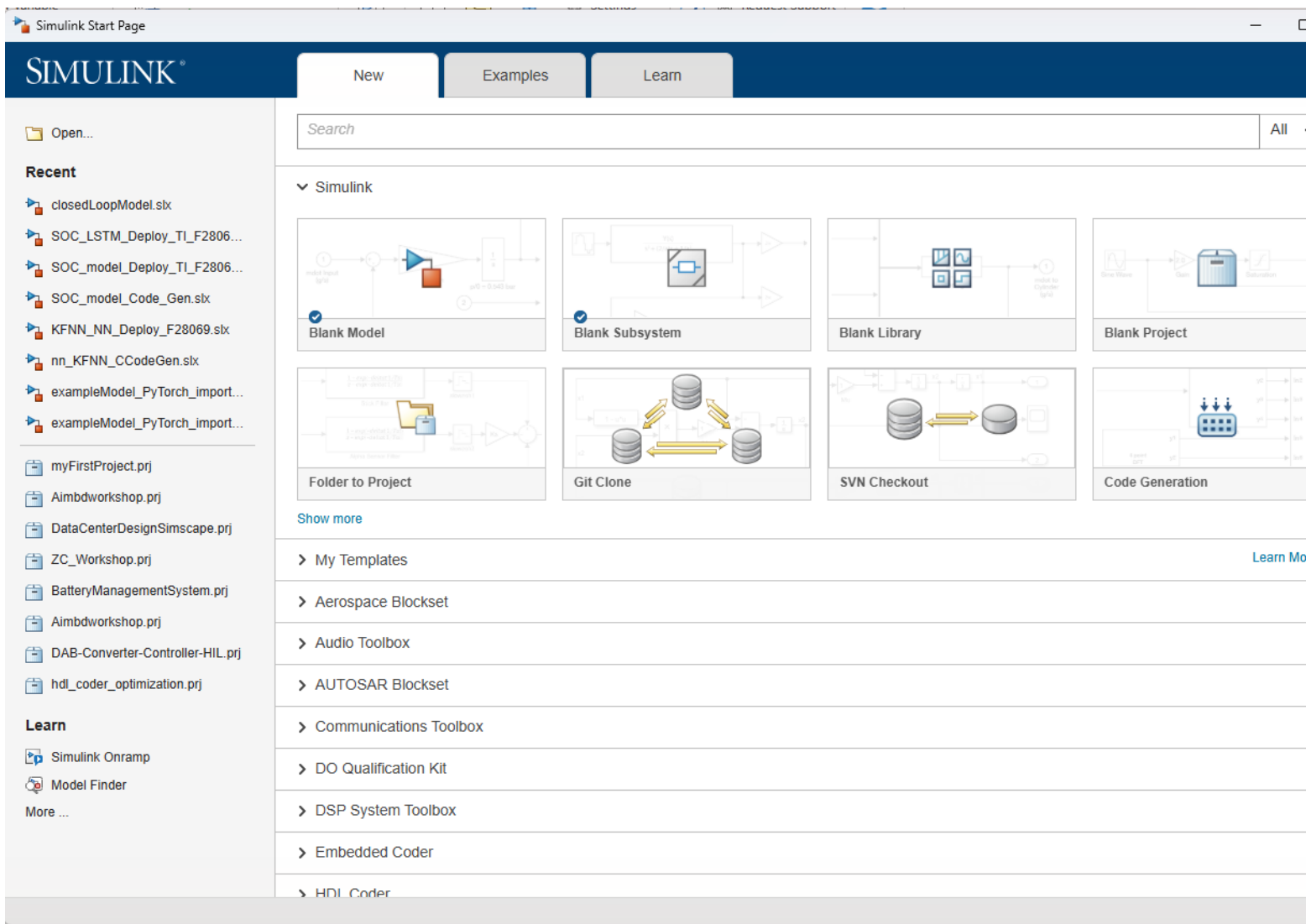


1. Opening Simulink - The Start Page

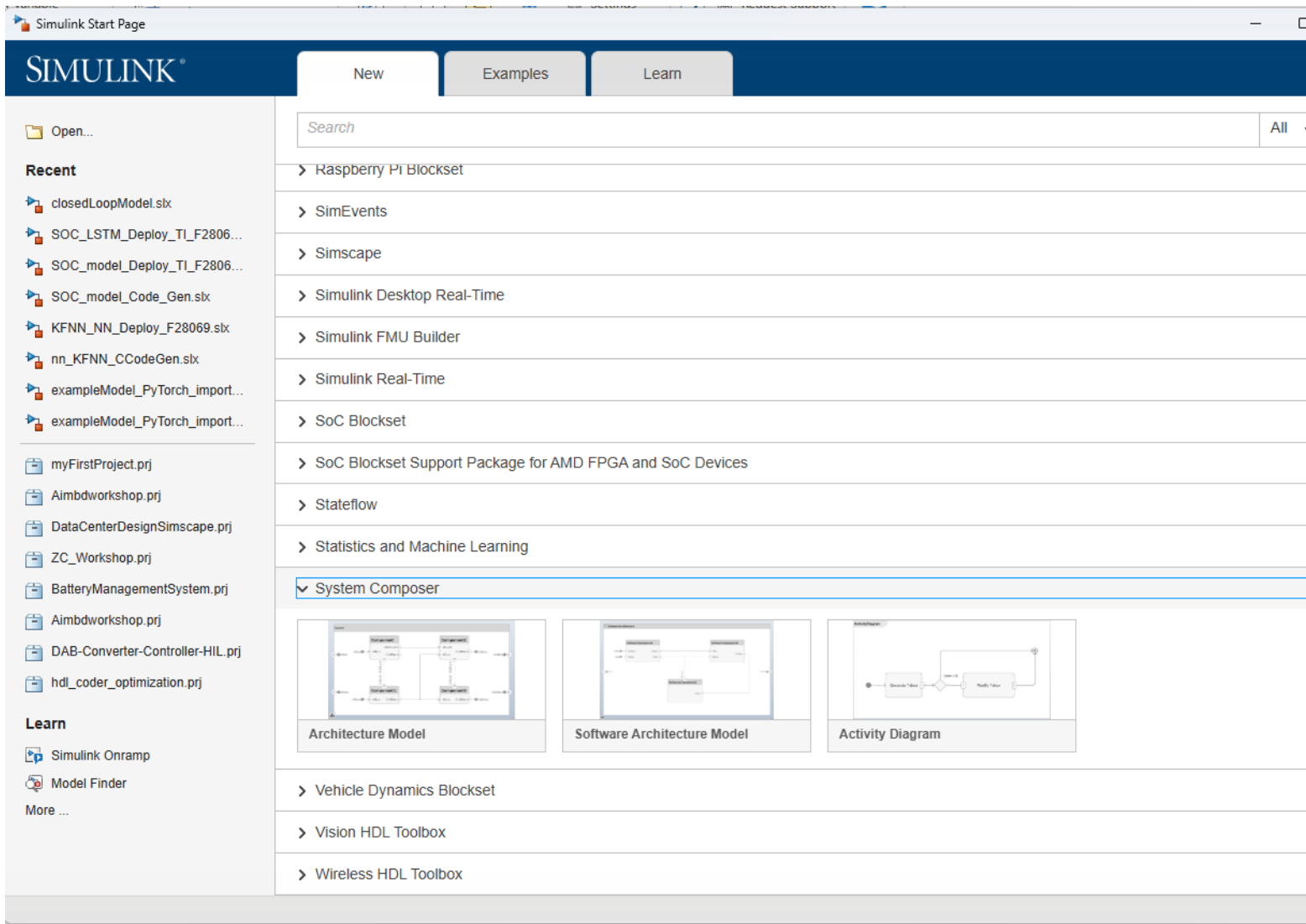
Launch Simulink from the Simulink button on the HOME tab of the MATLAB desktop, or type:

```
simulink
```

The Start Page appears. On the New tab you will find templates - Blank Model is the usual starting point. Recent files and projects are listed on the left.

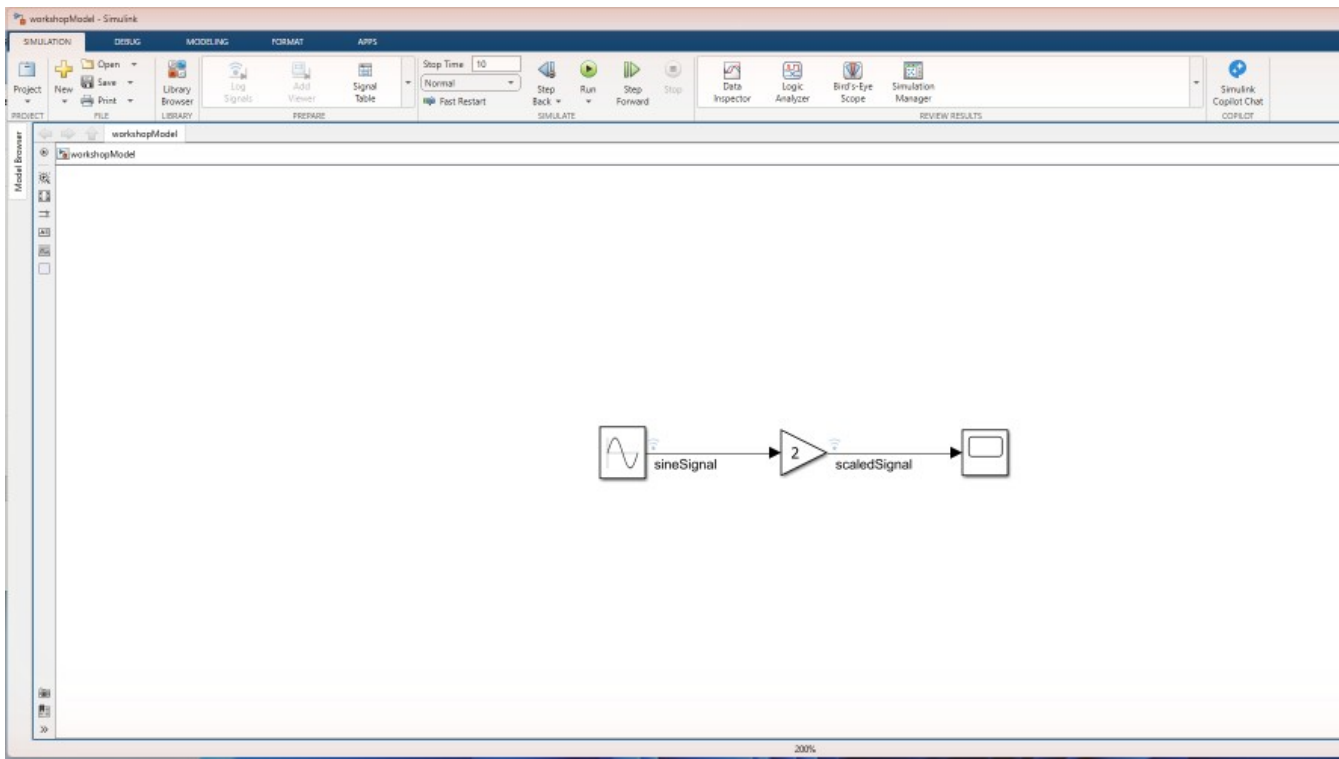


Every installed product adds its own section of templates. Scroll down to find System Composer - this is where you create Architecture Model, Software Architecture Model and Activity Diagram files for system-level design:



2. The Simulink Editor and Its Tabs

Open the demo model by double-clicking workshopModel.slx in the Project panel, or by typing:
`open_system("workshopModel")`

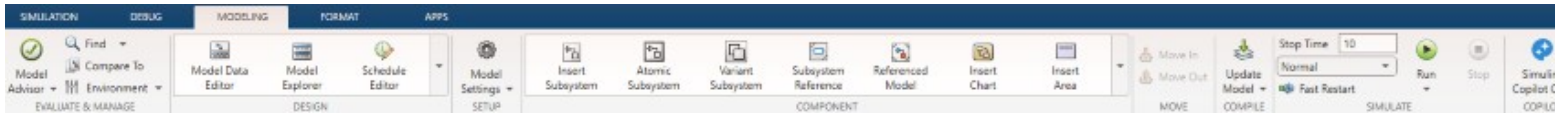


The toolbar tabs you will use most:

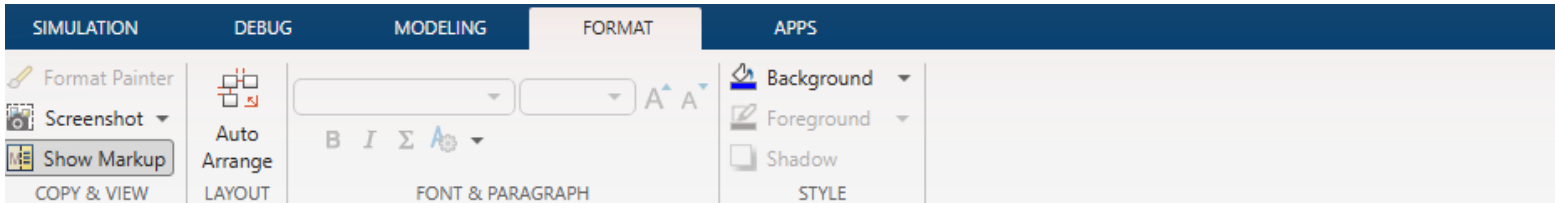
SIMULATION - the everyday workflow: open and save models, Library Browser, signal logging, simulation controls (Stop Time, mode, Run, Step) and review tools such as the Data Inspector:



MODELING - model-level tasks: Model Advisor checks, Model Explorer, Model Settings (solver and data import/export configuration), and inserting subsystems, charts or referenced models:

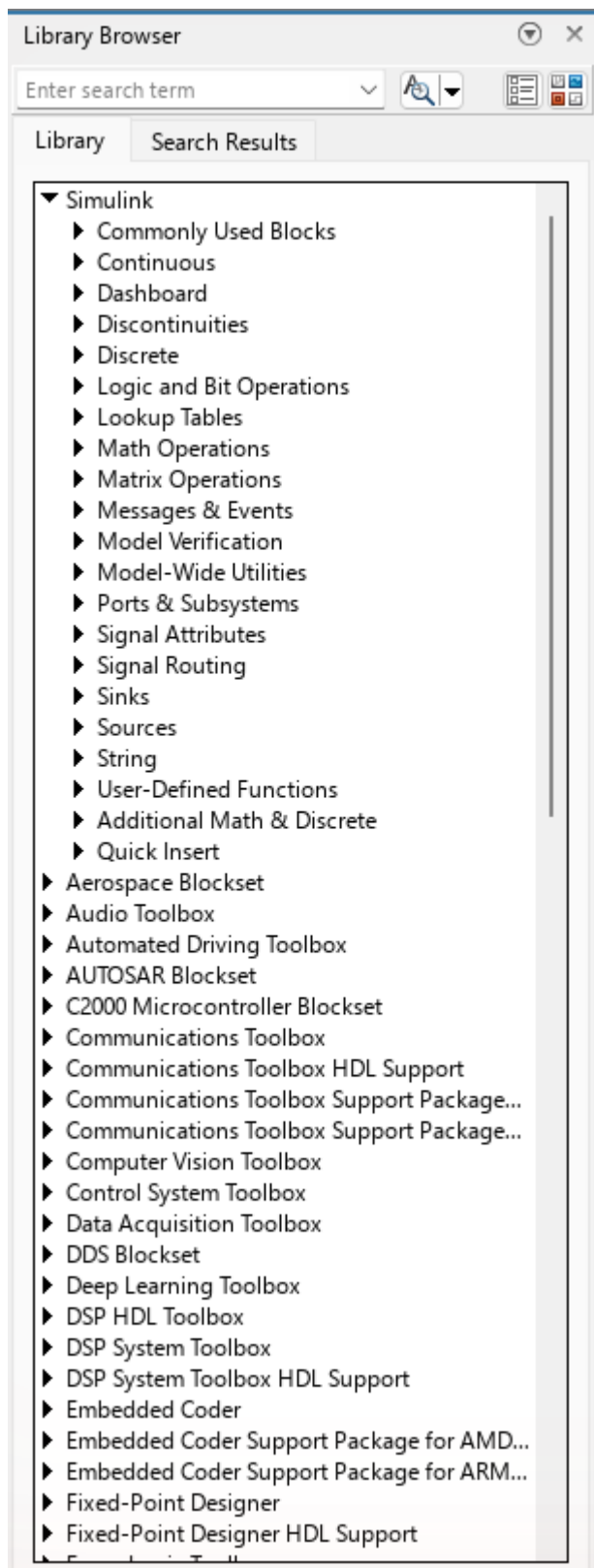


FORMAT - appearance: fonts, colors, alignment and Auto Arrange:

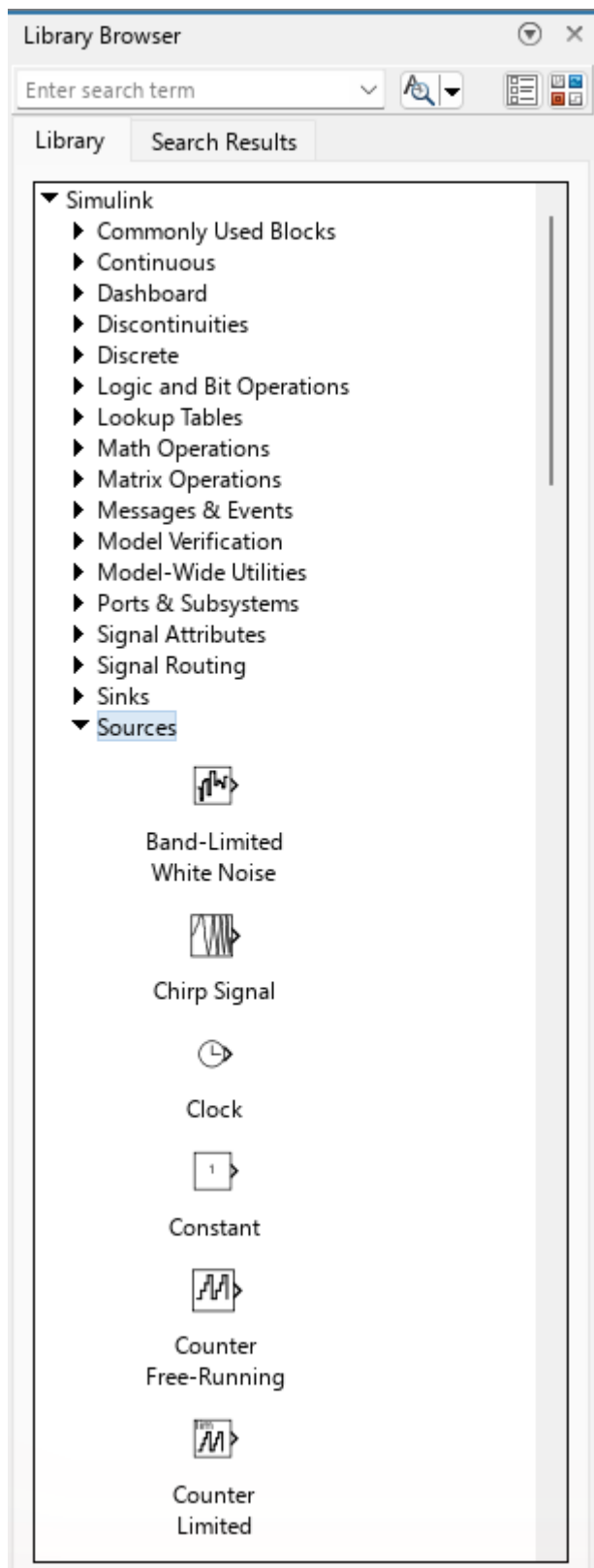


3. The Library Browser

The Library Browser (first button on the SIMULATION tab, or Ctrl+Shift+L) contains every block you can drag into a model, organized by product. Expand the core Simulink library to see its categories:



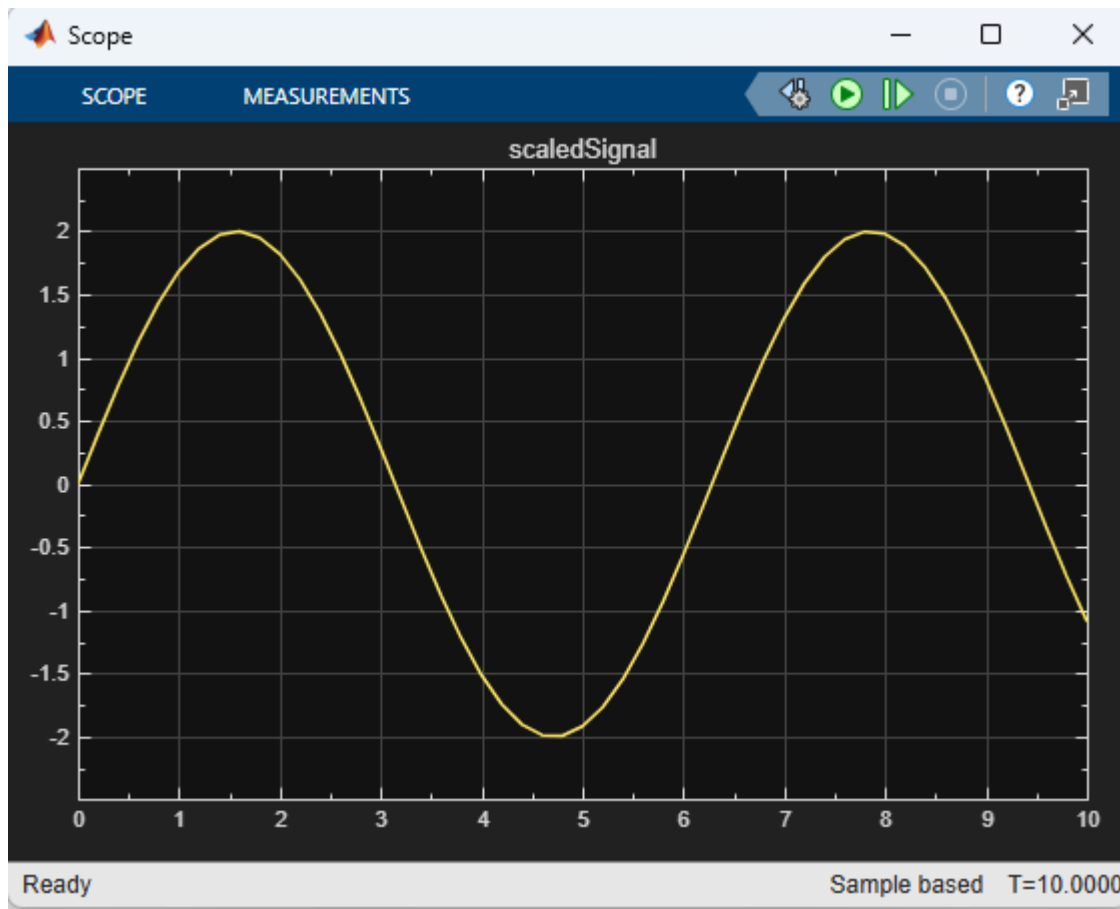
Expanding a category, such as Sources, shows the blocks it contains. Drag any block onto the canvas to use it:



Tip: you can also double-click on an empty spot of the canvas and type a block name for quick insertion.

4. Running a Simulation

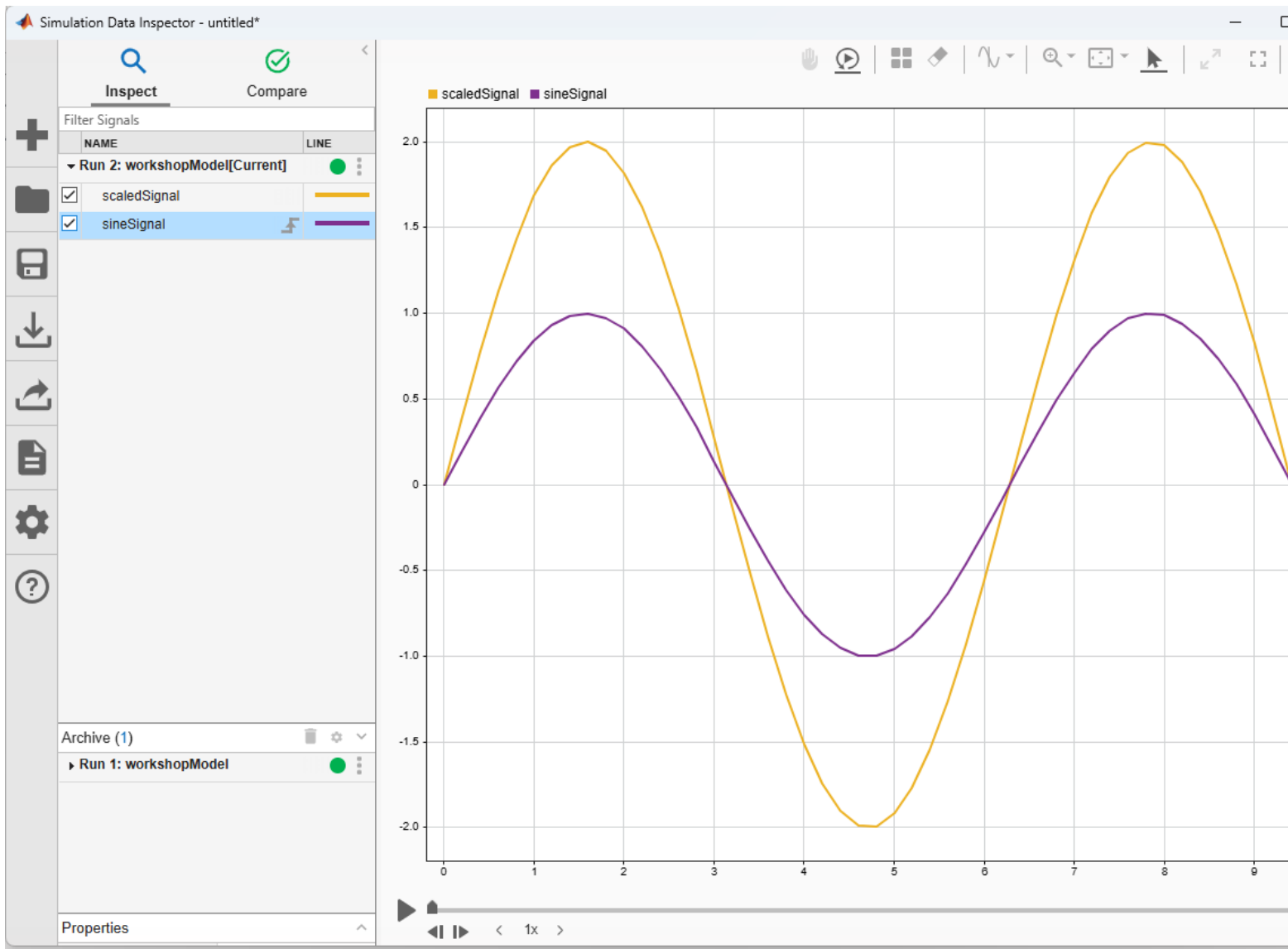
workshopModel.slx generates a sine wave, scales it by a factor of 2, and displays the result on a Scope. Set the simulation Stop Time (10 seconds here) in the SIMULATION tab, then press Run. Double-click the Scope block to see the result:



5. Simulation Data Inspector (SDI)

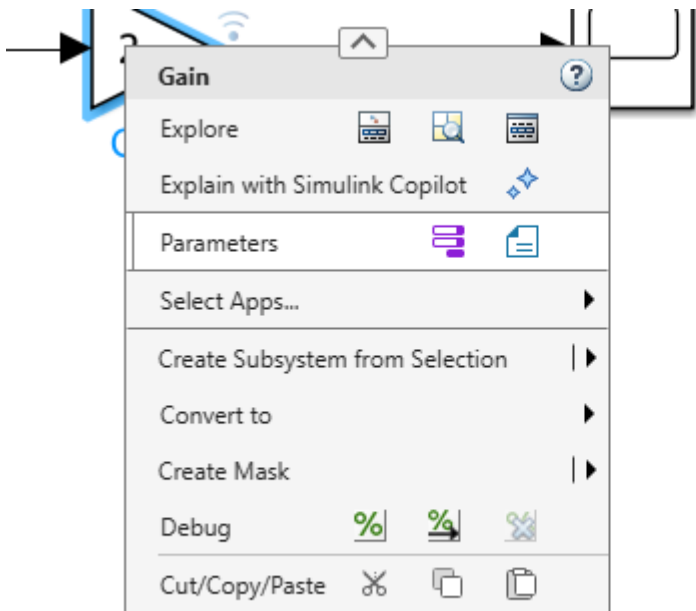
A Scope only shows what it is wired to. The Simulation Data Inspector collects every logged signal from every run, so you can browse and compare results without rewiring or re-running. To mark a signal for logging, right-click it and choose Log Selected Signals - a small antenna badge appears on the signal line. After simulating, open the SDI with the Data Inspector button on the SIMULATION tab.

Both signals in our model are logged. Tick them in the signal list to plot; each new simulation appears as a run, and the Compare tab can diff two runs:

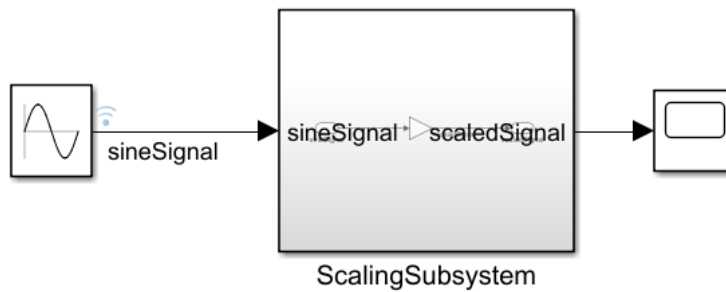


6. Subsystems

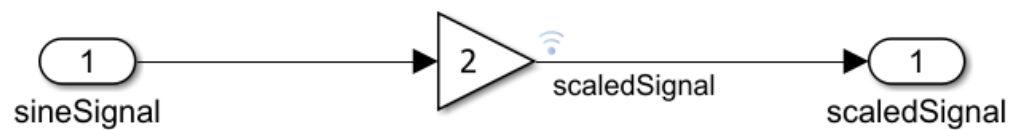
Subsystems group blocks into a single reusable unit and keep large diagrams readable. Select the blocks to group, right-click, and choose Create Subsystem from Selection (or press Ctrl+G):



The selection is replaced by a single subsystem block - here renamed to ScalingSubsystem:



Double-click the subsystem to look inside. Inport and Outport blocks were added automatically at the boundaries:



Use the breadcrumb at the top of the canvas to navigate back up the model hierarchy:



